PhySci/MiMG/CaSB M178

Homework 1

Due: 10/6/25 at 2:00PM PST

Notes: This homework involves performing simulations of the Posttranslational Modification (PTM) cycles and cascades we've been discussing in the first two class meetings. In the same folder on Bruinlearn where you obtained this document, you will also find a file called "HW1 template.ipynb" that contains a template Jupyter notebook that you can use as a starting point to complete the questions below. Please modify this notebook and use it as the starting point for answering the following problems.

To submit your homework, please answer the questions below. Note that you will have to **paste in several graphs** that you generate using the Jupyter notebook. After completing the questions, **save this document as a PDF** and upload it to Gradescope. **You must also upload the Jupyter notebook to Bruinlearn**. Finally, do not forget to **MARK YOUR QUESTIONS** when uploading to Gradescope.

Part 1: Protein Phosphorylation

In class, we talked about a kinase binding and phosphorylating a substrate protein "S." Just to re-iterate, the chemical equations of the system are the following:

$$\mathrm{K} + \mathrm{S} \xrightarrow{k_+} \mathrm{KS}$$

$$\mathrm{KS} \xrightarrow{k_-} \mathrm{K} + \mathrm{S}$$

$$\mathrm{KS} \xrightarrow{k_{cat}} \mathrm{K} + \mathrm{S}^*$$

where K and S are the free kinase and substrate, KS is the kinase-substrate complex, and S^* is the phosphorylated substrate. The various k values represent the rate parameters for each reaction.

In a posttranslational modification cycle, we also have a phosphatase enzyme, P, that binds to the phosphorylated substrate and undoes the modification. Its chemical reactions are:

$$\mathrm{P} + \mathrm{S}^* \xrightarrow{k_+} \mathrm{PS}^*$$

$$\mathrm{PS}^* \xrightarrow{k_-} \mathrm{P} + \mathrm{S}^*$$

$$\mathrm{PS}^* \xrightarrow{k_{cat}} \mathrm{P} + \mathrm{S}$$

which are very similar to the ones for the kinase, but P binds S* and produces the unmodified substrate S. Note that, for the purposes of this homework, we will use the same rate parameters for K and P.

In class, we wrote down the change equations for the simple case where there is just a kinase (i.e. no phosphatase). Please refer to the posted lecture notes if you need to refresh your memory for those change equations. Since the kinase and phosphatase work together on the same substrate, the more physiologically relevant situation involves having all three proteins (K, P and S) present in the same system.

For your reference, we have provided the change equations for the combined system below, which we worked out in class.

$$S' = -k_+ S \cdot K + k_- KS + k_{cat} PS^*$$

$$KS' = k_+ S \cdot K - k_- KS - k_{cat} KS$$

$$S^{*\prime} = -k_+ S^* \cdot P + k_- PS^* + k_{cat} KS$$

$$PS^{*\prime} = k_+ S^* \cdot P - k_- PS^* - k_{cat} PS^*$$

$$K' = -k_+ S \cdot K + k_- KS + k_{cat} KS$$

$$P' = -k_+ S^* \cdot P + k_- PS^* + k_{cat} PS^*$$

In the provided template Jupyter notebook, we have already coded up the change equations for this system. They are towards the top of the template.

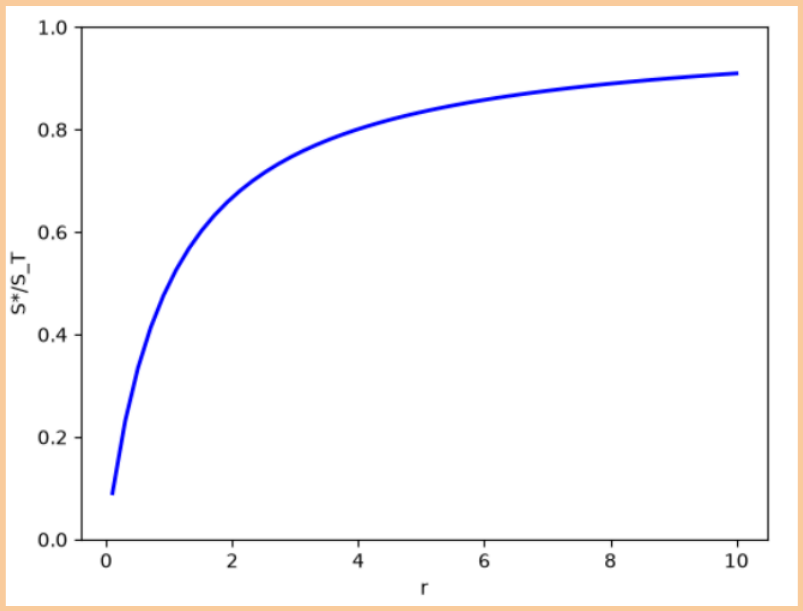
You will see that below the change equations there is a section called initial conditions. You will need to modify that section to add initial conditions for all six state variables. Note that the units of all the state variables are “micro Molar” (μM). You won’t have to explicitly set the units in the template, but it is good to know what they are. Set the initial conditions to: S(0) = 0.1, KS(0) = 0, S*(0) = 0, PS*(0) = 0, and P(0) =0.001. Note that this is an “unsaturated” case.

Below the section on initial conditions, there is a section of code that is called “PLOT Steady State Results.” This section of code will take the initial condition defined above and vary the initial value for K(0) along a specified range of possible K(0) values. For each K(0) value, it will simulate the model and find the fraction of the total Protein (S)

present that is phosphorylated. After doing this, it generates a plot of the steady-state value of S^*/S_T as a function of the parameter "r", as described in lecture.

1. (3 points) Run the code blocks listed in the description above and paste the steady state result curve in the space below.

-

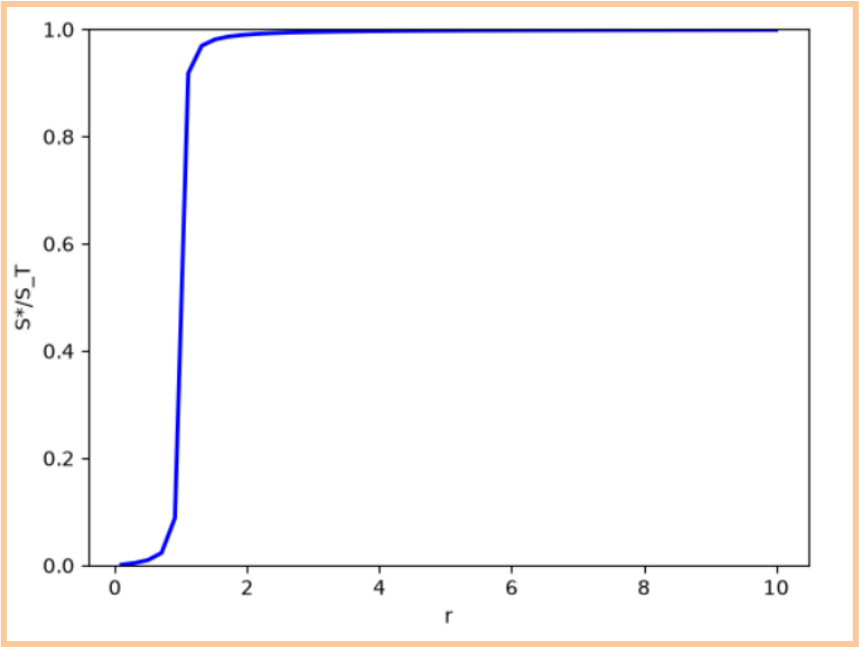


2. (6 Points) Would you describe the behavior as switch like or not? Explain in your own words in the space below.

- Answer: I would not describe this behavior as switch-like at all. As our value increases, the fraction of phosphorylated substrate has a gradual increase rather than an abrupt change over a small r range that would be descriptive of a switch. It being smooth shows that the system is not acting as ultrasensitive.

3. (3 Points) Modify the initial conditions so $S(0)$ = 1,000, and re run the code that calculates and plots the S^*/S_T vs. *r* graph. (Note that this is the "saturated" case). Paste your graph below.

-



4. (6 Points) Would you describe the saturated case as having switch like behavior? Explain in your own words in the space below.

- Answer: I would describe this saturated case as having a switch like behavior. The fraction of phosphorylated substrate has sharp changes over a small range of r which is indicative of a switch. The quick response means that small changes in input can produce large changes in output, so the system is ultrasensitive, much more compared to the unsaturated system.

One of the more unrealistic things about the PTM cycle model described above is that there is no synthesis and degradation of the substrate protein. In real cells, proteins experience turnover; they are not just made one time by the cell and then sit around forever. Real proteins can get damaged, unfold, or for other reasons "go bad." So, the cell constantly degrades proteins to keep damaged proteins from causing problems, and to make up for this, it is also constantly synthesizing proteins to make up for the proteins that are lost.

To illustrate how we might model this turnover process, consider some generic protein in the cell called "Z." Let's say the cell makes this protein at a constant rate Q and degrades it at a per-molecule rate of δ. We can write this as chemical reactions in the following way:

$$\xrightarrow{Q} \mathrm{Z}$$

$$\mathrm{Z} \xrightarrow{\delta}$$

The resulting change equation for Z would be:

$$Z' = Q - \delta Z$$

Let's make a model of the PTM cycle where the substrate is synthesized and degraded. To do so, use the following system of chemical reactions:

$$\begin{aligned}
\mathrm{K} + \mathrm{S} &\xrightarrow{k_+} \mathrm{KS} \\
\mathrm{KS} &\xrightarrow{k_-} \mathrm{K} + \mathrm{S} \\
\mathrm{KS} &\xrightarrow{k_{cat}} \mathrm{K} + \mathrm{S}^* \\
\mathrm{P} + \mathrm{S}^* &\xrightarrow{k_+} \mathrm{PS}^* \\
\mathrm{PS}^* &\xrightarrow{k_-} \mathrm{P} + \mathrm{S}^* \\
\mathrm{PS}^* &\xrightarrow{k_{cat}} \mathrm{P} + \mathrm{S} \\
&\xrightarrow{Q} \mathrm{S} \\
\mathrm{S} &\xrightarrow{\delta} \\
\mathrm{S}^* &\xrightarrow{\delta} \\
\mathrm{KS} &\xrightarrow{\delta} \mathrm{K} \\
\mathrm{PS}^* &\xrightarrow{\delta} \mathrm{P}
\end{aligned}$$

We know this looks complicated, but notice that the top six reactions are just the standard PTM loop.

5. (6 Points) In the space below, write out the modified change equations that include production and degradation. Note that you should have an equation for $S', K', KS', S^{*'}, P'$ and $PS^{*'}$.

5 Answer:

S_prime = -k_plus*S*K + k_minus*KS + k_cat*PSstar + Q - delta*S

KS_prime = k_plus*S*K - k_minus*KS - k_cat*KS - delta*KS

Sstar_prime = -k_plus*Sstar*P + k_minus*PSstar + k_cat*KS - delta*Sstar

PSstar_prime = k_plus*Sstar*P - k_minus*PSstar - k_cat*PSstar - delta*PSstar

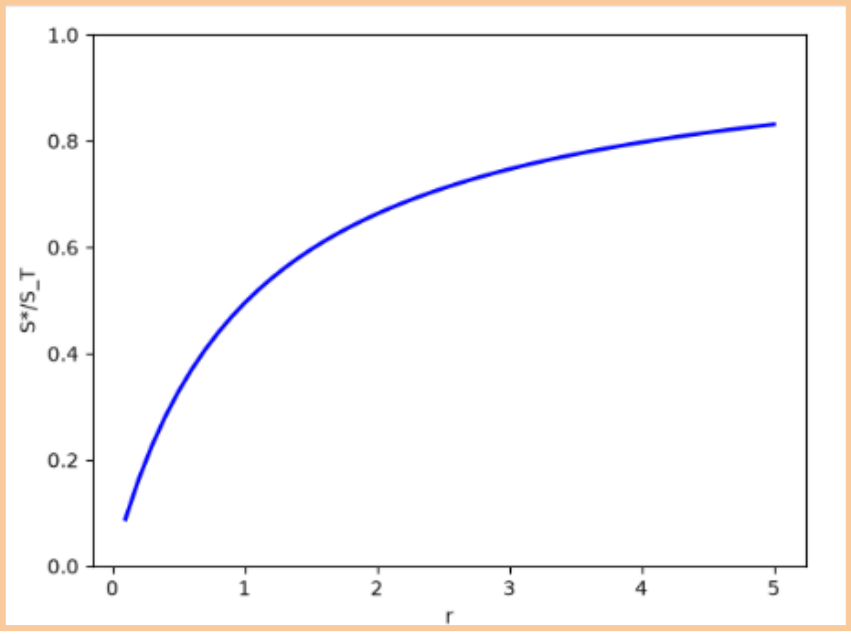
K_prime = -k_plus*S*K + k_minus*KS + k_cat*KS + delta*KS

P_prime = -k_plus*Sstar*P + k_minus*PSstar + k_cat*PSstar + delta*PSstar

To simulate this system, look below where you were working in the Jupyter notebook template for the section labeled “Second model: includes synthesis and degradation.” We will be looking at the steady-state response of this for the two different conditions that we’ve been discussing. Doing this is a little more complicated than the case without synthesis and degradation. Note that the rate parameters k_+, $k-$, and k_{cat} don’t change here. We have to add the parameter δ, which has already been done for you, and that won’t change between the saturated and unsaturated cases. But the value of the parameter Q will change, so be sure to watch for that!
Focus first on the unsaturated case. You will see that again there is code in the template for generating the steady-state response curve. First, run that code with the following initial conditions: S(0) = 0.1, KS(0) = 0, S*(0) = 0, PS*(0) = 0,and P(0) =0.001 (these are the default values in the template). We will also set Q to be δ * S(0). Run the code for this case.

6. (3 Points) In the space below, paste your steady state response curve for the unsaturated condition described above. The code for this section should be found in the section named: *“STEADY STATE responses of the second model.”*

-

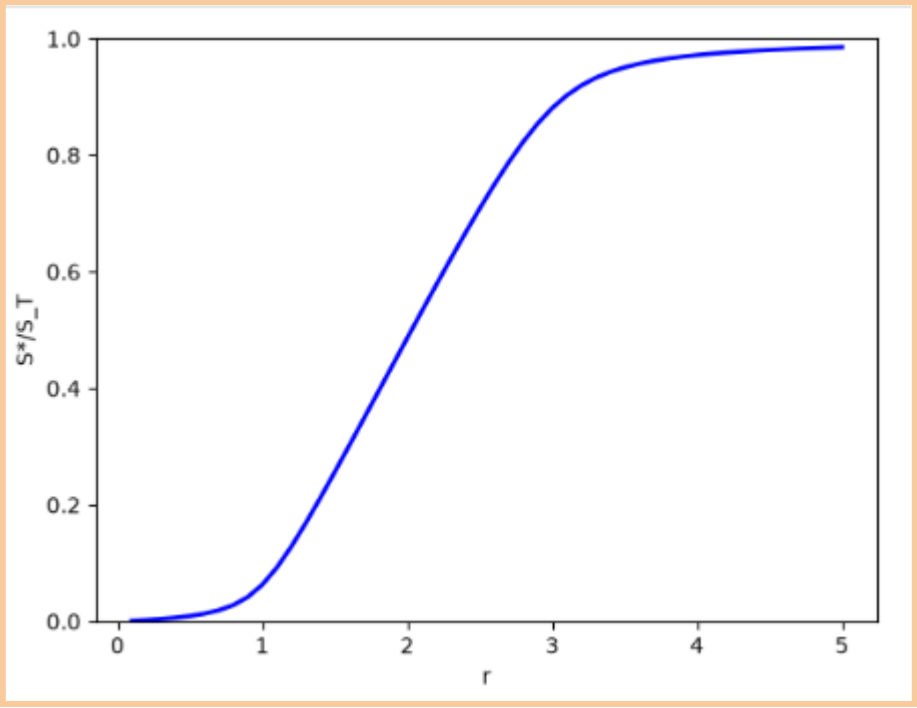


7. (3 Points) How does the unsaturated result of the model with synthesis and degradation compare to the unsaturated result of the model without synthesis or degradation? Which one is more ultrasensitive?

- 7 Answer: The output of the unsaturated model with synthesis and degradation looks very similar to the output of the model without synthesis and degradation. Both of the outputs have gradually, non switch-like increases. However, the model without synthesis and degradation does appear to be slightly more ultrasensitive as its response curve is a bit steeper.

Now let's run the saturated case. Here, set the initial conditions to: $S(0) = 1000$, $KS(0) = 0$, $S^*(0) = 0$, $PS^*(0) = 0$ and $P(0) = 0.001$, and set Q to $\delta^* S(0)$.

8. (3 Points) Modify your code from Exercise 6 to simulate the saturated condition and paste steady state response curve that you generate in the space below.



9. (**3 Points)** How does the saturated result of the model that with synthesis and degradation compare to the saturated result of the model without synthesis or degradation? Which one is more ultrasensitive?

- 9 Answer: The saturated model with synthesis and degradation appears less ultrasensitive compared to the saturated model without synthesis and degradation. With the model w/o synthesis and degradation the responses change very sharply and abruptly over a very small range of r, within r less than 2. With synthesis and degradation, the curve is more gradual, and spread over a wider range of r. So the model without synthesis and degradation is more ultra sensitive.

10. (6 Points) You have now considered two models: one without synthesis/degradation and one that includes synthesis/degradation. Between the two models tested, which one is more biologically relevant? Answer this question and explain why in the space below.

- 10 Answer: The model that includes synthesis and degradation would be more biologically relevant as in real cells, proteins are continuously synthesized and degraded. Proteins can become damaged, unfold, or need to be replaced at certain points in their life span. By including this process the model becomes more realistic than assuming that the substrate protein remains in the system forever.

Part 2: Kinase Cascades

In class, we also talked about a kinase cascade, such as the MAPK signaling pathway. Here is an example model with three layers:

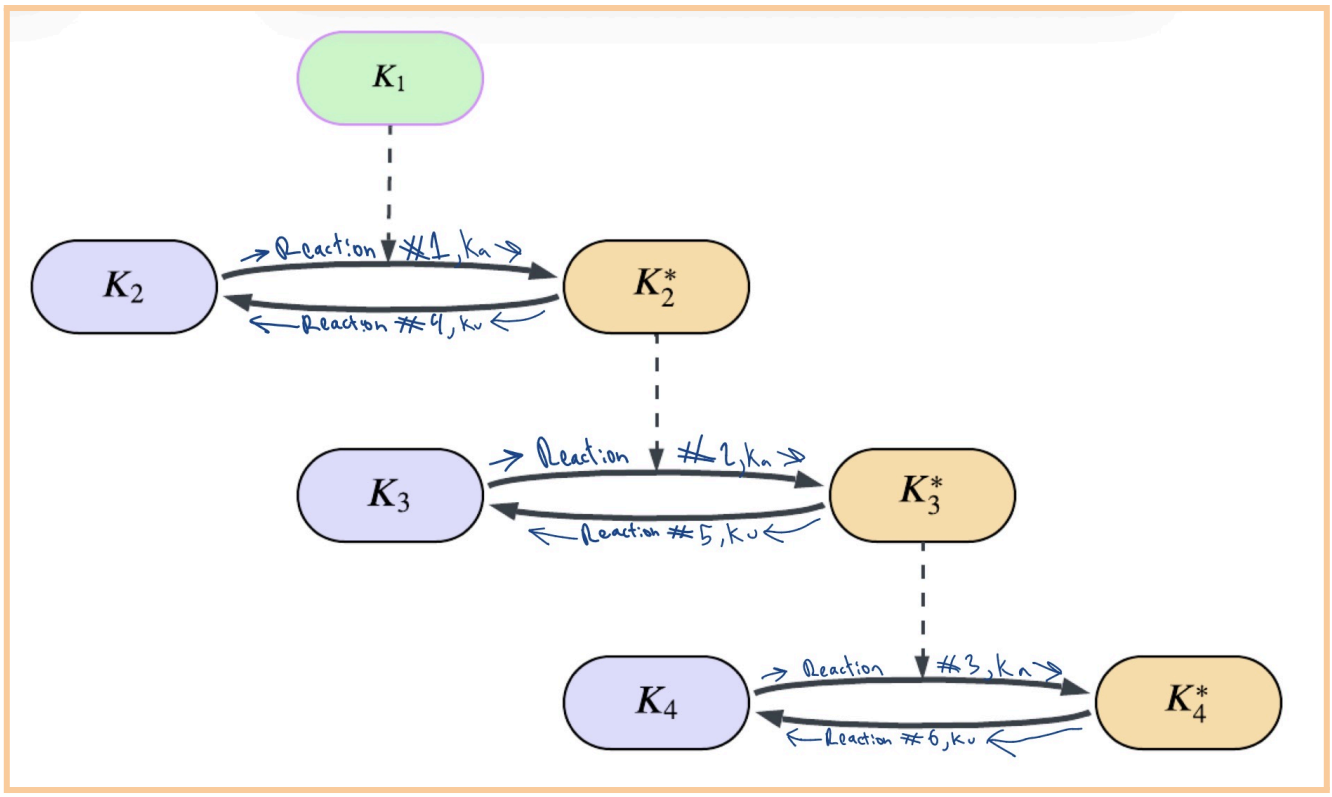
$$(1) \quad K_1^* + K_2 \xrightarrow{k_a} K_1^* + K_2^*$$
$$(2) \quad K_2^* + K_3 \xrightarrow{k_a} K_2^* + K_3^*$$
$$(3) \quad K_3^* + K_4 \xrightarrow{k_a} K_3^* + K_4^*$$

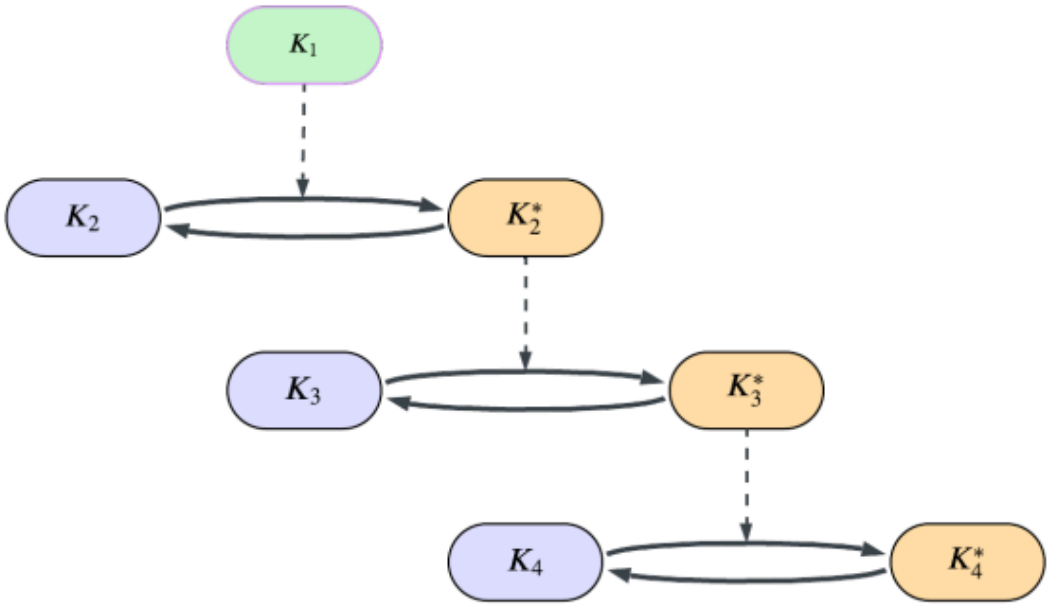
Upon a receptor binding event at the cell surface, the first kinase species, K_1, is changed (phosphorylated) to its active form, K_1^* (not modeled here). This active kinase species then activates the next kinase species, K_2, and so on.

The active kinase species also experience deactivation (dephosphorylation) modeled as follows:

$$(4) \quad K_2^* \xrightarrow{k_u} K_2$$
$$(5) \quad K_3^* \xrightarrow{k_u} K_3$$
$$(6) \quad K_4^* \xrightarrow{k_u} K_4$$

11. (6 Points) Below is an example of a **chemical reaction network**, that visually describes the 3-layer cascade. Annotate each arrow with the correct chemical equation (using the numbers 1-6 above) and the rate constant that describes it.





We can write the change equations like we discussed in class for this system:

$$
\begin{aligned}
K_2' &= -k_a K_1^* K_2 + k_u K_2^* \\
K_2^{*\prime} &= k_a K_1^* K_2 - k_u K_2^* \\
K_3' &= -k_a K_2^* K_3 + k_u K_3^* \\
K_3^{*\prime} &= k_a K_2^* K_3 - k_u K_3^* \\
K_4' &= -k_a K_3^* K_4 + k_u K_4^* \\
K_4^{*\prime} &= k_a K_3^* K_4 - k_u K_4^*
\end{aligned}
$$

In the template Jupyter notebook we provided, we have already coded up the change equations for this system. They are in the Cascade model section and are titled, "3 layer model", "4 layer model", and "5 layer model". While the 3 layer model has been implemented for you, the code for the other two models needs to be modified to accurately simulate the system.

To convert the "3 layer model" to the 4 layer model, we need to add the following two new chemical reactions:

$$(7) \quad K_4^* + K_5 \xrightarrow{k_a} K_4^* + K_5^*$$

$$(8) \quad K_5^* \xrightarrow{k_u} K_5$$

12. (3 Points) Add these change equations to the Jupyter notebook in the appropriately labelled areas to complete the "4-layer model". Take a screenshot of your modified code for this section and paste it in the space below.

```
# Below, ADD the additional change equations you found to complete the four layer model
###
K5_prime = -k_a*K4star*K5 + k_u*K5star
K5star_prime = k_a*K4star*K5 - k_u*K5star
###
```

To convert the "4 layer model" to the 5 layer model, we need to add the following two new chemical reactions:

$$(9) \quad K_5^* + K_6 \xrightarrow{k_a} K_5^* + K_6^*$$

$$(10) \quad K_6^* \xrightarrow{k_u} K_6$$

13. (3 Points) Add these changes to the “4-layer model” to make the “5-layer model” and write down the change equations for the “5-layer model” in the corresponding location in the Jupyter notebook. Once finished, take a screenshot of your code and paste it below.

```
# 5 layer model
# This is the definition of our system of ODEs for the model
def five_layer_model(states,t):
    # Our "states" list has the state variables in the following order:
    # K2, K2star, K3, K3star, K4, K4star, K5, K5star, K6, K6star
    K2 = states[0]
    K2star = states[1]
    K3 = states[2]
    K3star = states[3]
    K4 = states[4]
    K4star = states[5]
    K5 = states[6]
    K5star = states[7]
    K6 = states[8]
    K6star = states[9]
    # The change equations here are the same as above for the three layer model
    K2_prime = -k_a*K1star*K2 + k_u*K2star
    K2star_prime = k_a*K1star*K2 - k_u*K2star
    K3_prime = -k_a*K2star*K3 + k_u*K3star
    K3star_prime = k_a*K2star*K3 - k_u*K3star
    K4_prime = -k_a*K3star*K4 + k_u*K4star
    K4star_prime = k_a*K3star*K4 - k_u*K4star

    # Below, ADD the additional change equations you found to complete the five layer model
    #(note you will need to include the additions you made for the four layer model as well here)
    ###
    K5_prime = -k_a*K4star*K5 + k_u*K5star
    K5star_prime = k_a*K4star*K5 - k_u*K5star
    K6_prime = -k_a*K5star*K6 + k_u*K6star
    K6star_prime = k_a*K5star*K6 - k_u*K6star
    ###

    states_prime = [K2_prime,K2star_prime,K3_prime,K3star_prime,K4_prime,K4star_prime,K5_prime,K5star_prime,K6_prime,K6star_prime]
    return states_prime
```

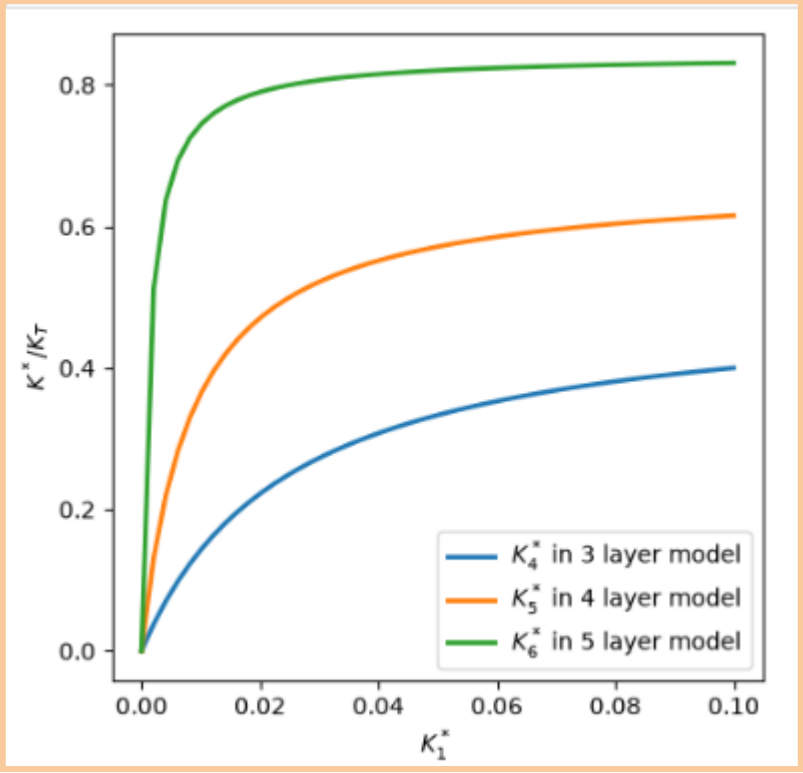

```
# Below, ADD the additional change equations you found to complete the five layer model
#(note you will need to include the additions you made for the four layer model as well here)
###
K5_prime = -k_a*K4star*K5 + k_u*K5star
K5star_prime = k_a*K4star*K5 - k_u*K5star
K6_prime = -k_a*K5star*K6 + k_u*K6star
K6star_prime = k_a*K5star*K6 - k_u*K6star
###
```

You will see that there is a section called "Cascades: STEADY-STATE responses". Here we will vary the amount of K_1^*, the activating signal, in the system and observe how it changes the steady state amount of final kinase activation for our different sized cascades.

Note the initial conditions (shared for the 3, 4, and 5 layer models) are set at the beginning of this section of code such that all of the downstream kinases are in the inactive form at the start. Furthermore, moving down the layers, the total amount of kinase protein doubles.

Below that, there is a section of code that is called "Kinase Cascades: PLOT your steady-state results" Note that this section of the code changes the initial value of K_1^*, from 0 to 0.1 (double its default value of 0.05). It then runs a simulation of the 3, 4, and 5 layer models with those parameters, and saves the result of the final time point. After doing this, it generates a plot of the steady-state value of $K_{\#}^*/K_{\#total}$ as a function of the parameter K_1^*. Here # refers to the final kinase in each cascade (# = 4 for the 3-layer model, # = 5 for the 4-layer model, # = 6 for the 5-layer model).

14. (3 Points) Run the sections of code described in the previous section and paste the resulting graph below.



15. (6 Points) How does the amount of K_1, the activating signal, needed to achieve maximum activation of the final level kinase vary with the number of levels in the cascade? How then might a kinase cascade provide "signal amplification"?

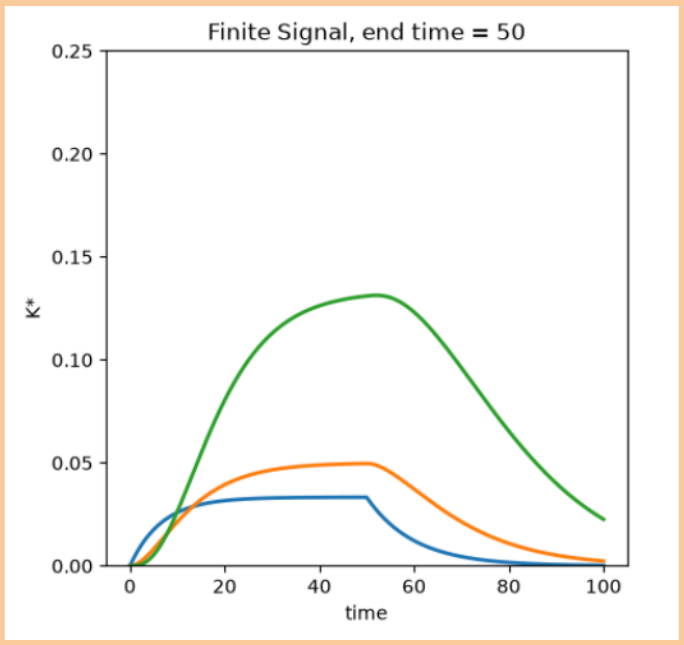
- 15 Answer: When the number of levels of cascades increases, we need less K1* to produce a large activation of the final kinase. This means that longer cascades are actually able to amplify the original signal. Small amounts of activating signal at the first level are able to lead to much larger response at the final because each activated kinase is able to activate molecules in the next layer of the cascade.

In all the models we considered above, the external signal was held at a *constant* value and we looked at the steady-state output of the system. In a real biological context, incoming signals are often *transient*, in that they are present only for a particular period of time.

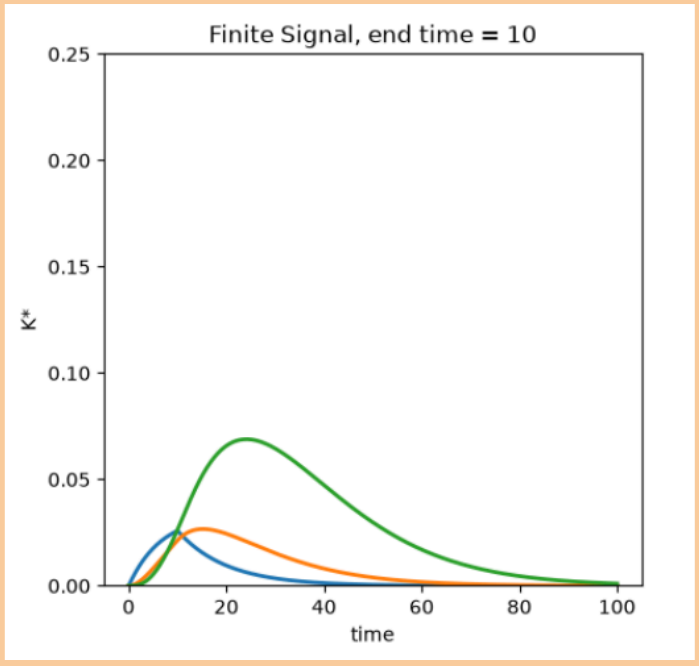
We will now consider how this kind of transient signal might impact a kinase cascade. Underneath the section where you plot the kinase activation over time, the code next defines a new model, "DYNAMICS: 3 layer model with finite signal", where K_1^*, the activating signal, can be turned off after some time. That time is called "end_signal". You will try two different values of end_signal and see how this influences the behavior of the system.

Note that, in these graphs, the X-axis is **TIME**, not the amount of incoming signal as in your previous plots!!

16. (3 Points) Run the code and paste your graph ("# PLOT dynamics of kinase activation with finite signal") and set end_signal = 50. Paste the resulting graph below.



17. (3 Points) Run the code and paste your graph ("# PLOT dynamics of kinase activation with finite signal") and set end_signal = 10. Paste the resulting graph below.



18. (1 Point) In which of the two cases you tried (the first with end_signal = 50 or the second with end_signal = 10) did the output signal (K_4^*) last the longest?

- 18 Answer: The output signal K4* lasted longer when end_signal = 50.

19. (6 Points) The above two graphs are for the same exact model, but for different durations of input signal. Explain why the results differ in terms of how *long* it takes for the final kinase in a cascade to be activated. Why would having the input signal "on" for longer have this effect?

- 19 Answer: There are differing results because the activation has to pass through levels of the kinase cascade before the final kinase is fully activated. This process creates a delay between the input signal and the response of the final kinase. When that input signal is left on for longer, the upstream kinases are active for more time, allowing more downstream kinase molecule activation, and as a result the final kinase can reach a higher level of activation. Its signal then lasts longer. When input is turned off early, there is less time for activation to propagate through the cascade, so the final response is much smaller and shorter lived.